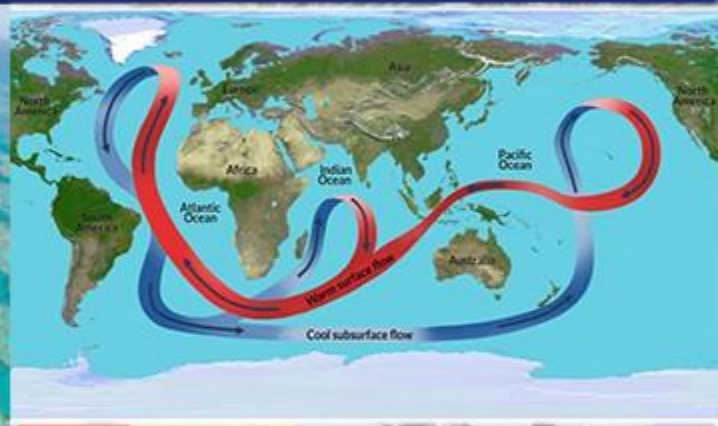


GEOGRAPHY

PHYSICAL GEOGRAPHY



OCEANOGRAPHY

MODULE - 1

**OCEAN BOTTOM
RELIEF**

OCEANOGRAPHY**TOPICS TO BE COVERED**

- 1. OCEAN BOTTOM RELIEF**
- 2. SALINITY AND TEMPERATURE OF OCEANS**
- 3. OCEAN TIDES**
- 4. OCEAN CURRENTS**
- 5. CORAL REEFS**



OCEANOGRAPHY

RELIEF OF THE OCEAN FLOOR



- 1 The oceans are confined to the great depressions of the earth's outer layer.
- 2 The oceans, unlike the continents , merge so naturally into one another that it is hard to demarcate them.
- 3 The geographers have divided the oceanic part of the earth into oceans, like the Pacific, the Atlantic, the Indian ocean etc.
- 4 The various seas, bays, gulfs and other inlets are parts of these large oceans.



OCEANOGRAPHY

RELIEF OF THE OCEAN FLOOR

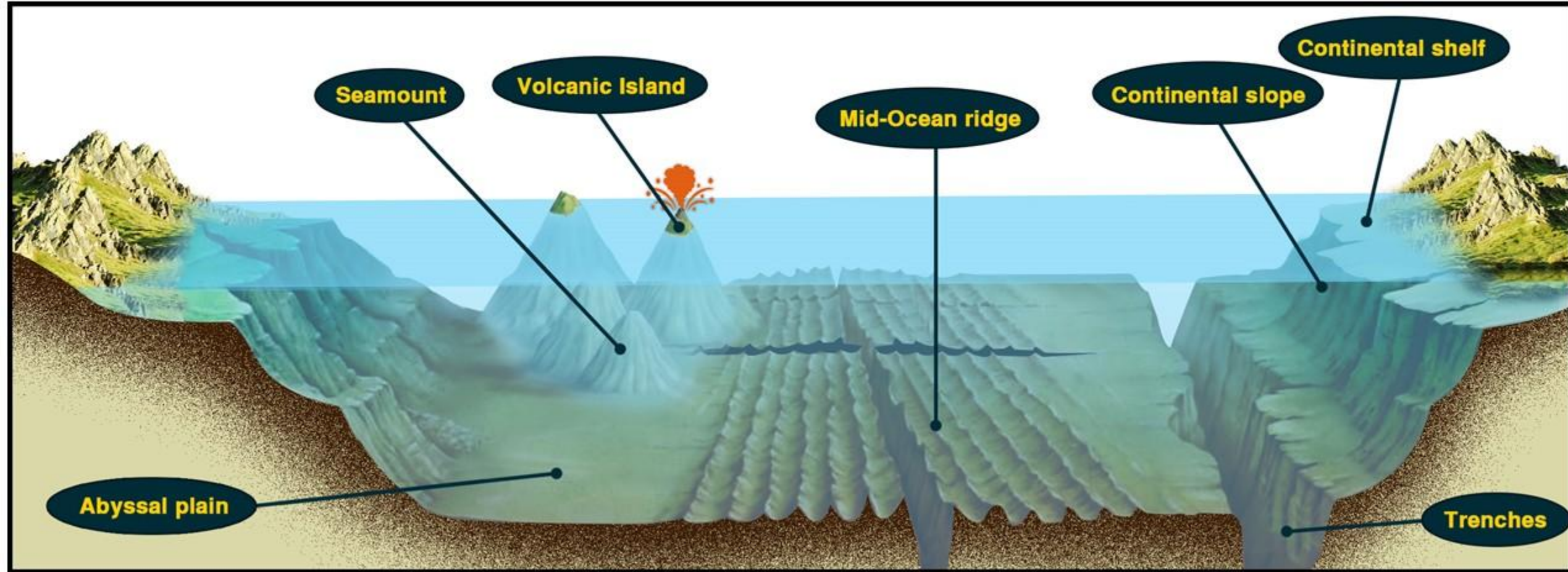


- 5 A major portion of the ocean floor is found between 3-6 km below the sea level.
- 6 The 'land' under the waters of the oceans, that is, the ocean floor exhibits complex and varied features as those observed over the land.
- 7 The floors of the oceans are rugged with the world's largest mountain ranges, deepest trenches and the largest plains.
- 8 These features are formed, like those of the continents, by the factors of tectonic, volcanic and depositional processes.



OCEANOGRAPHY

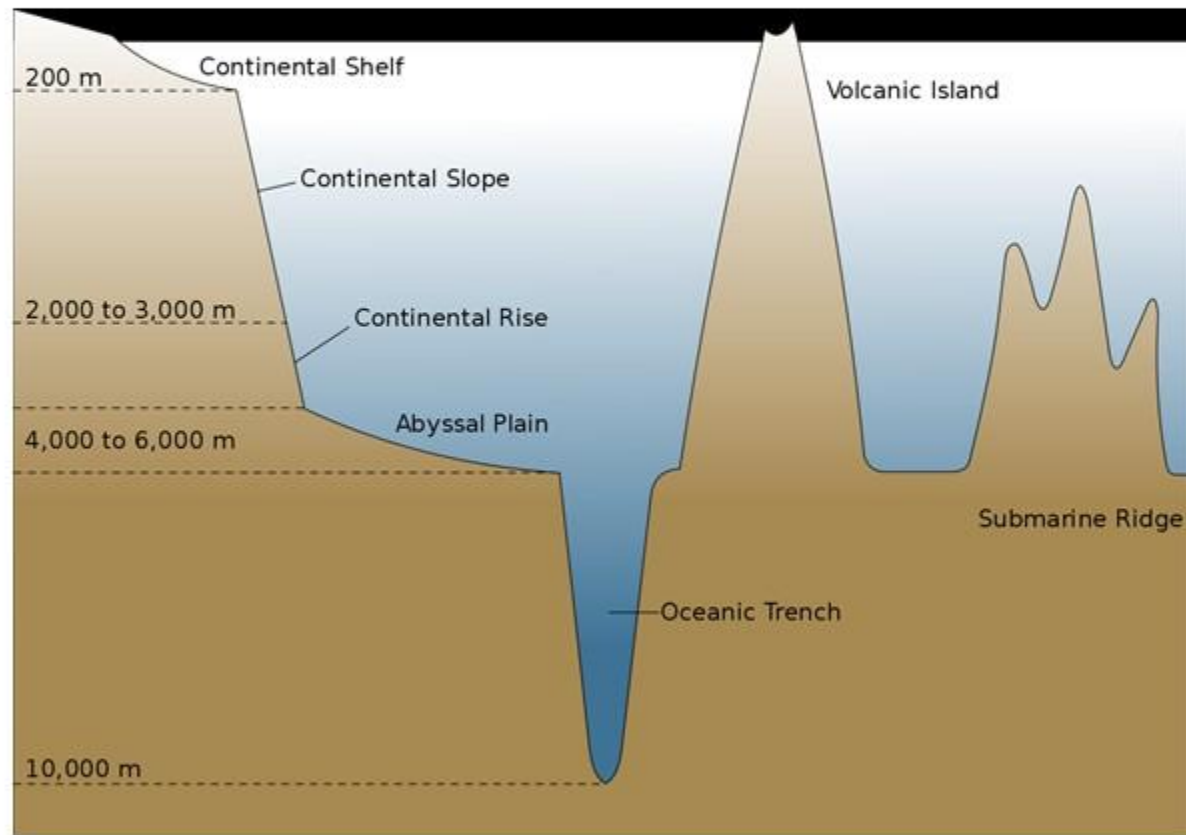
RELIEF OF THE OCEAN FLOOR





OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS



The ocean floors can be divided into four major divisions:

- i. The Continental Shelf
- ii. The Continental Slope
- iii. The Deep Sea Plain
- iv. The Oceanic Deeps.

Besides these divisions, there are also major and minor relief features in the ocean floors like ridges, hills, sea mounts, guyots, trenches, canyons, etc.

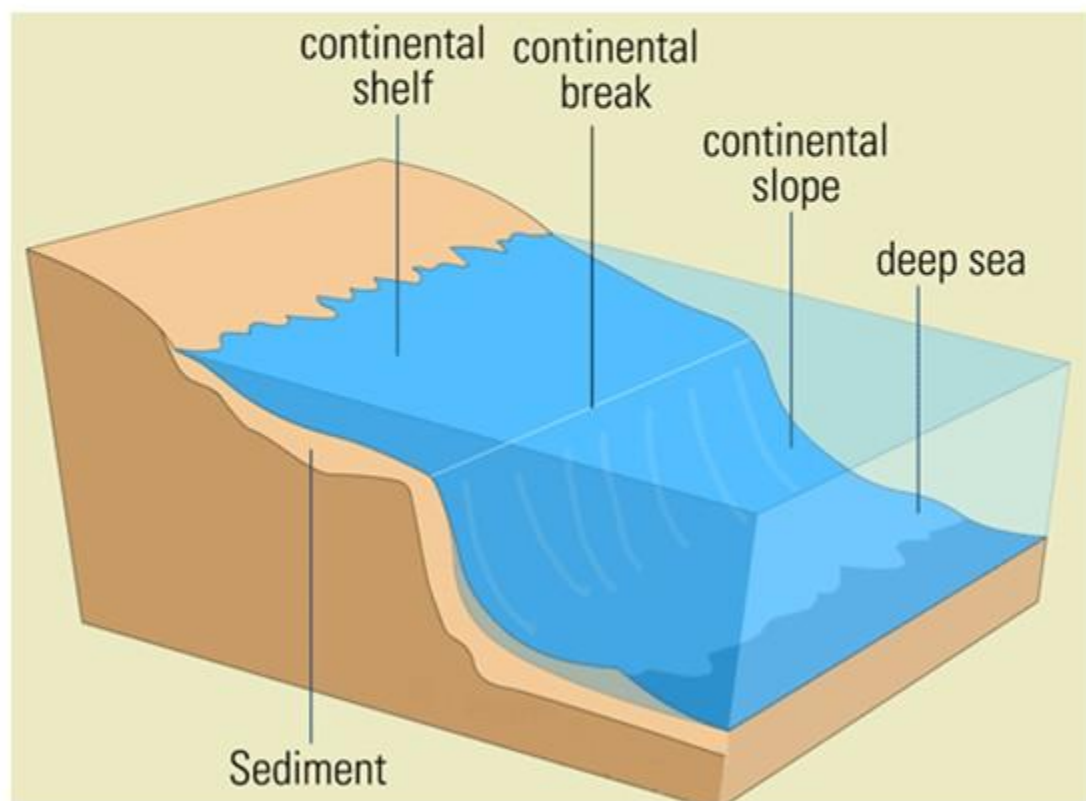


OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS



1 CONTINENTAL SHELF



- 1 The continental shelf is the extended margin of each continent occupied by relatively shallow seas.
- 2 It is the shallowest part of the ocean showing an average gradient of 1° or even less.
- 3 The shelf typically ends at a very steep slope, called the shelf break.
- 4 The width of the continental shelves vary from one ocean to another.
- 5 The average width of continental shelves is about 80 km.



OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

1 CONTINENTAL SHELF



6 The shelves are almost absent or very narrow along some of the margins like the coasts of Chile, the west coast of Sumatra, etc.

7 On the contrary, the Siberian shelf in the Arctic Ocean, the largest in the world, stretches to 1,500 km in width.

DIVISIONS OF THE OCEAN FLOORS

1 CONTINENTAL SHELF

SIBERIAN SHELF

Siberian shelf is subdivided into the Kara Shelf, the Laptev Shelf and the East Siberian Shelf.

Eastwards it merges into the Chukchi Shelf (of the Chukchi Sea) shared by Eurasia and North America (i.e., by Russia and the United States) and westwards it merges into the Barents Shelf of the Barents Sea.





OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

1 CONTINENTAL SHELF



8 The continental shelves are covered with variable thickness of sediments brought down by rivers, glaciers, wind, from the land and distributed by waves and currents.

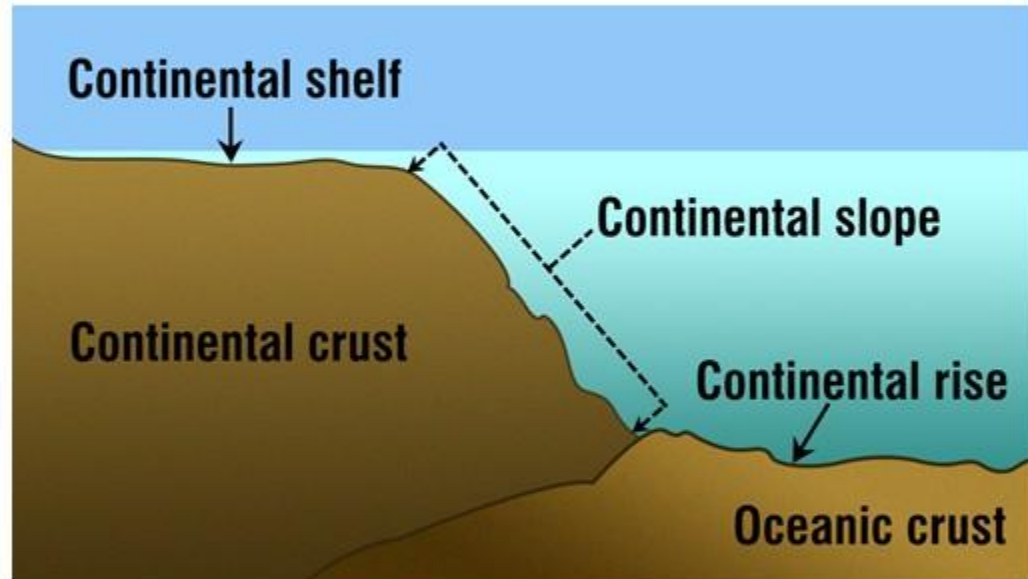
9 Massive sedimentary deposits received over a long time by the continental shelves, become the source of fossil fuels.



OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

2 CONTINENTAL SLOPE



- 1 The continental slope connects the continental shelf and the ocean basins or its floor.

- 2 It begins where the bottom of the continental shelf sharply drops off into a steep slope.

- 3 The gradient of the slope region varies between $2-5^\circ$.

- 4 The depth of the slope region varies between 200 and 3,000 m.

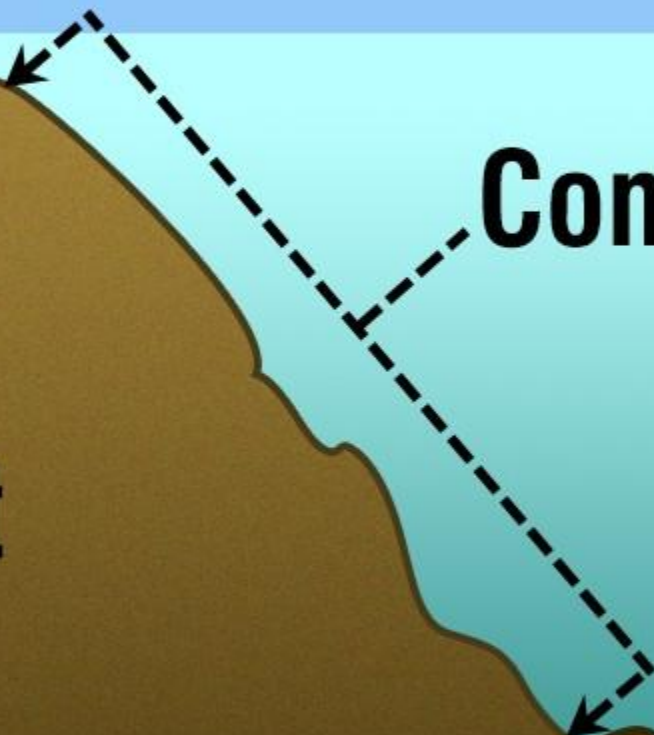
- 5 The slope boundary indicates the end of the continents.

- 6 Canyons and trenches are observed in this region.

Continental shelf



Continental slope

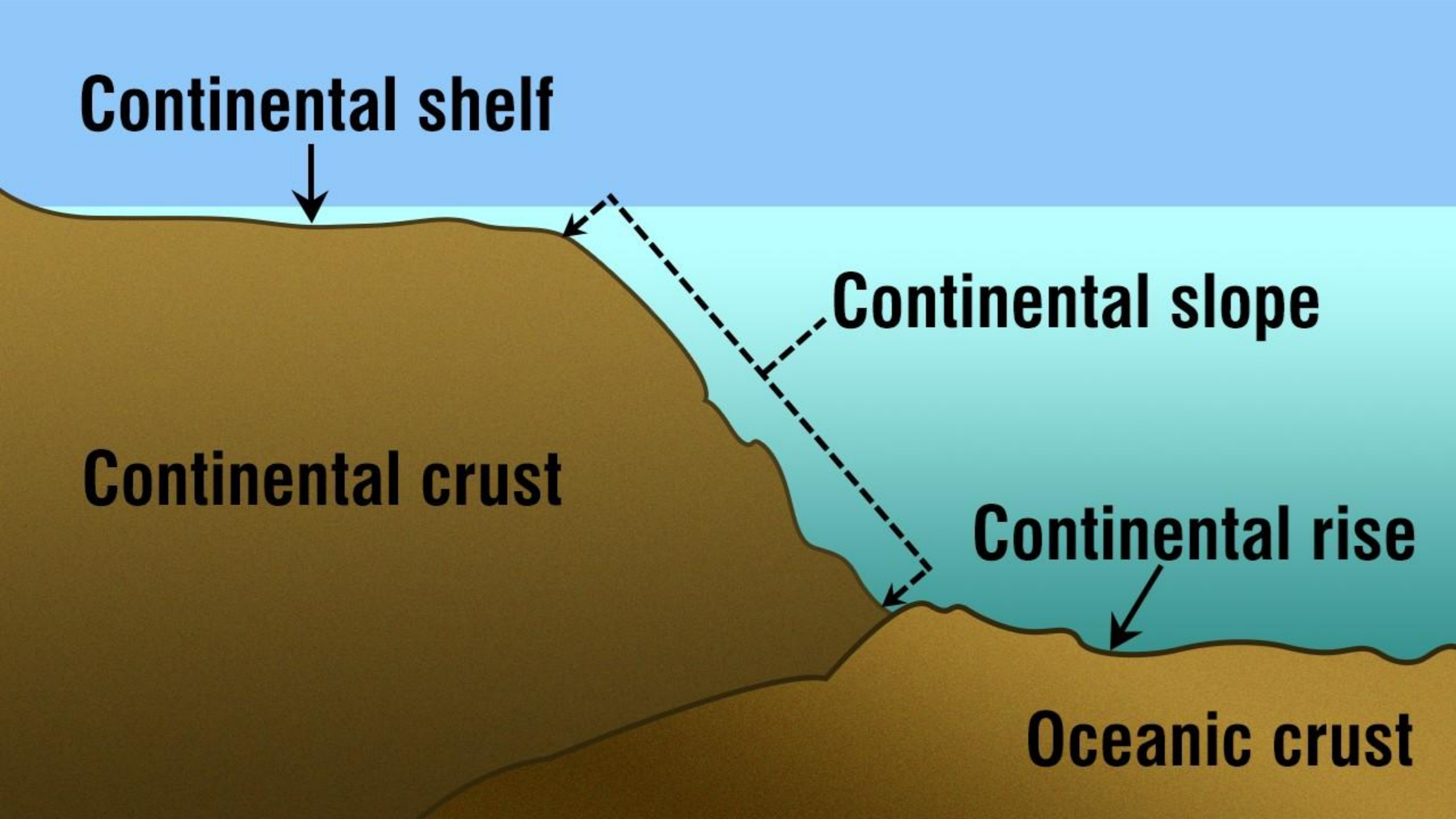


Continental crust

Continental rise



Oceanic crust



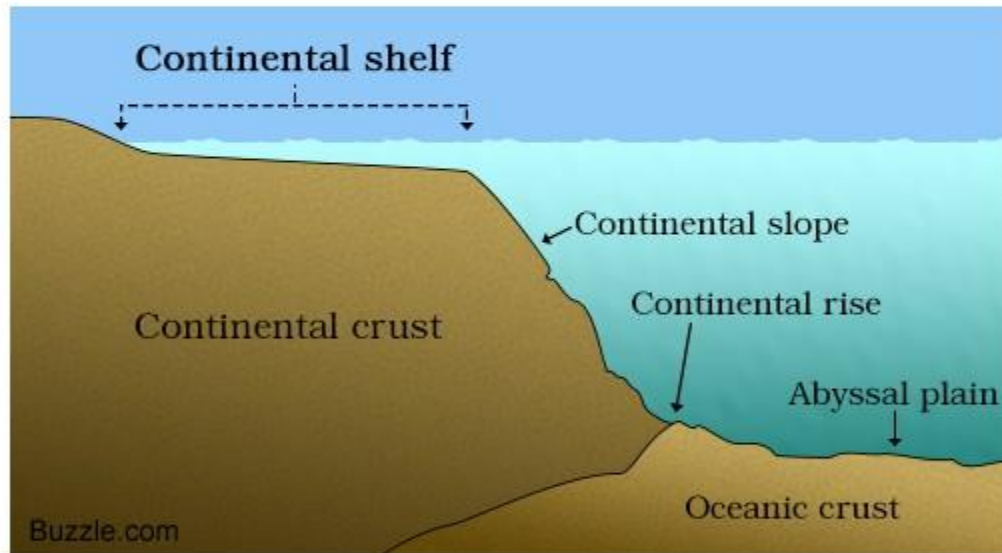


OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

3

DEEP SEA PLAIN



- 1 Deep sea plains are gently sloping areas of the ocean basins.
- 2 These are the flattest and smoothest regions of the world.
- 3 The depths vary between 3,000 and 6,000m.
- 4 These plains are covered with fine-grained sediments like clay and silt.

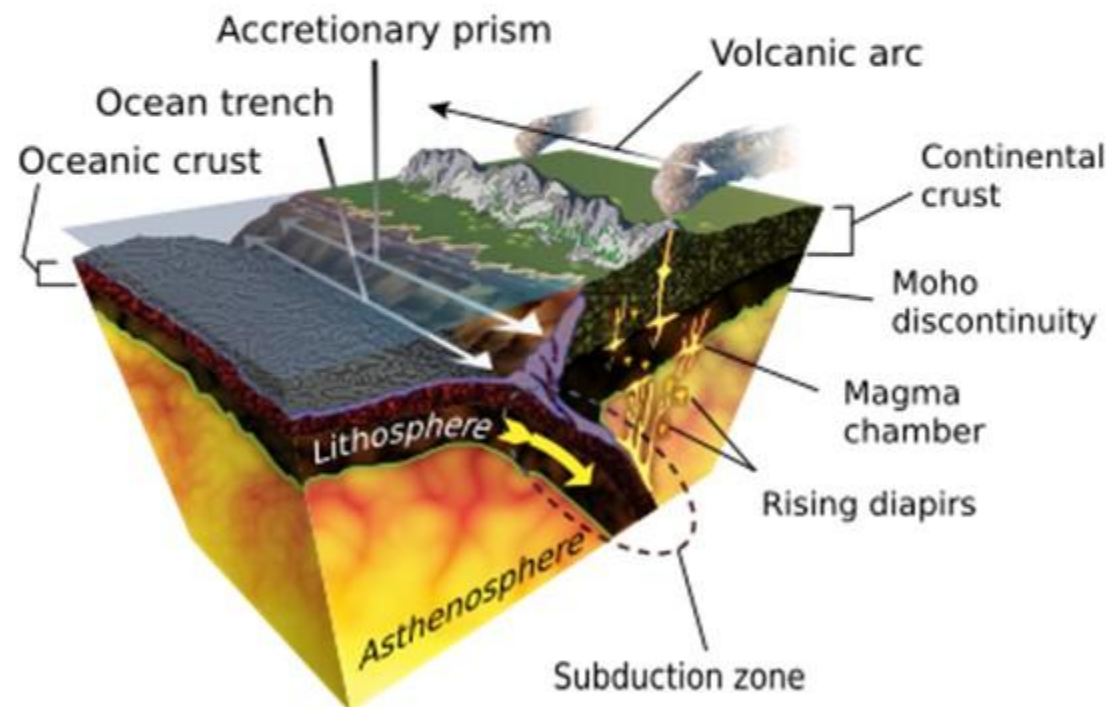


OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

4

OCEANIC DEEPS OR TRENCHES



- 1 These areas are the **deepest parts** of the oceans.
- 2 The trenches are **relatively steep sided, narrow basins**.
- 3 They are some **3-5 km deeper** than the surrounding ocean floor.
- 4 They occur at the **bases of continental slopes** and along island arcs and are associated with active volcanoes and strong earthquakes.
- 5 That is why they are very significant in the study of plate movements.

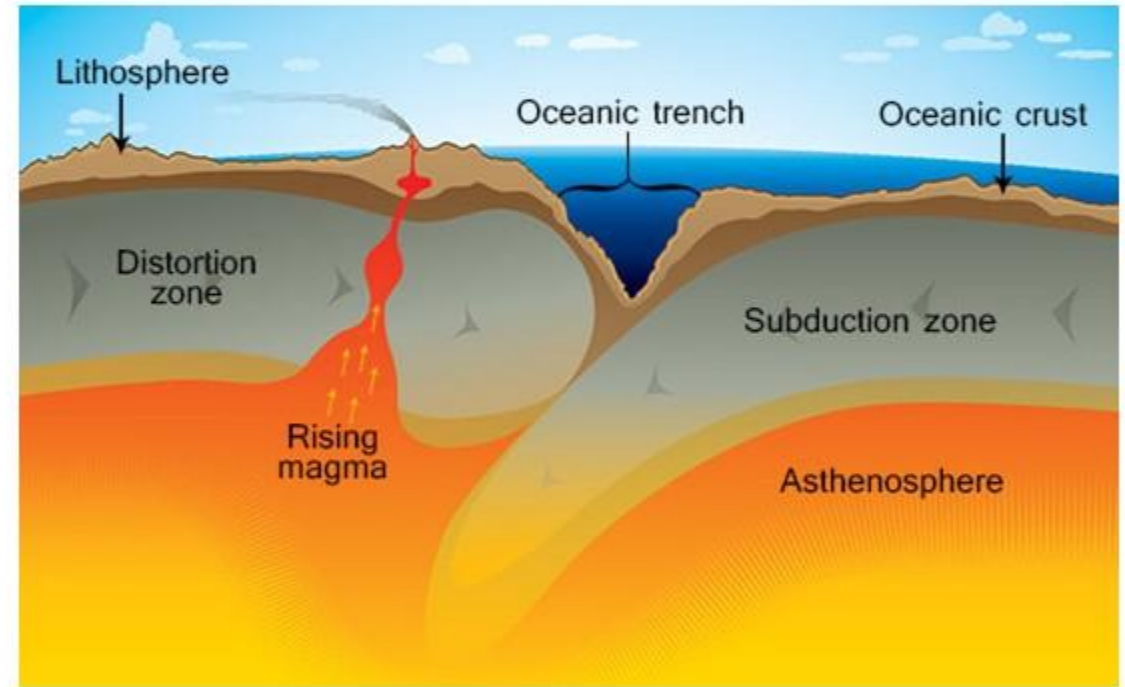
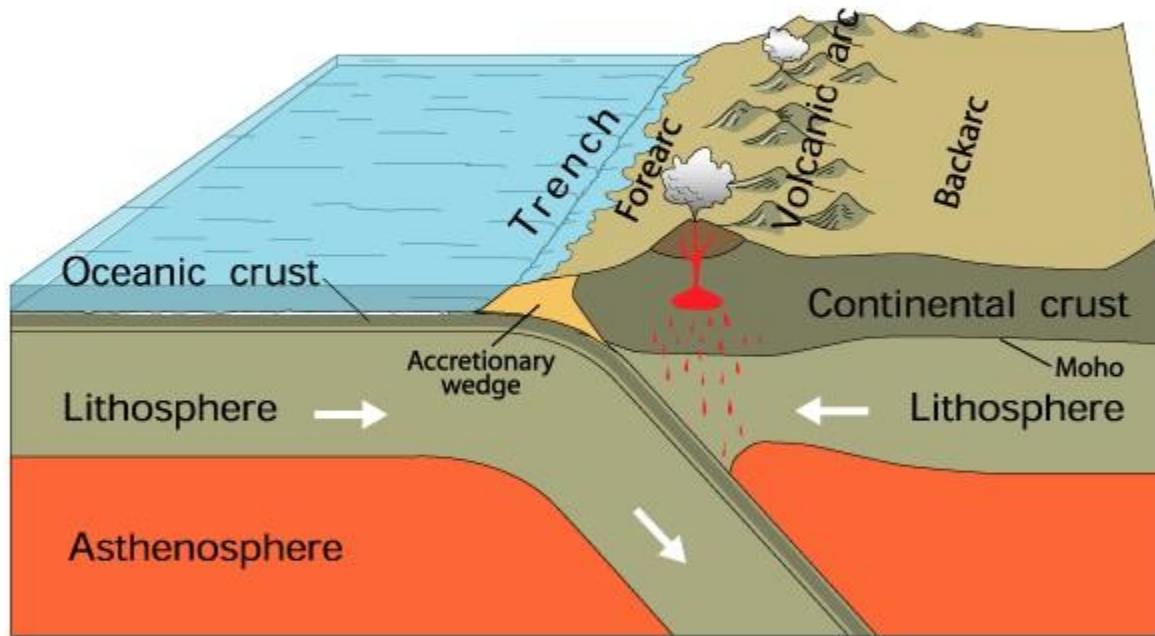


OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

4

OCEANIC DEEPS OR TRENCHES



Ocean trenches are steep depressions in the deepest parts of the ocean where an old ocean crust from one tectonic plate is pushed beneath another plate, raising mountains, causing earthquakes, and forming volcanoes on the seafloor and on land.



OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

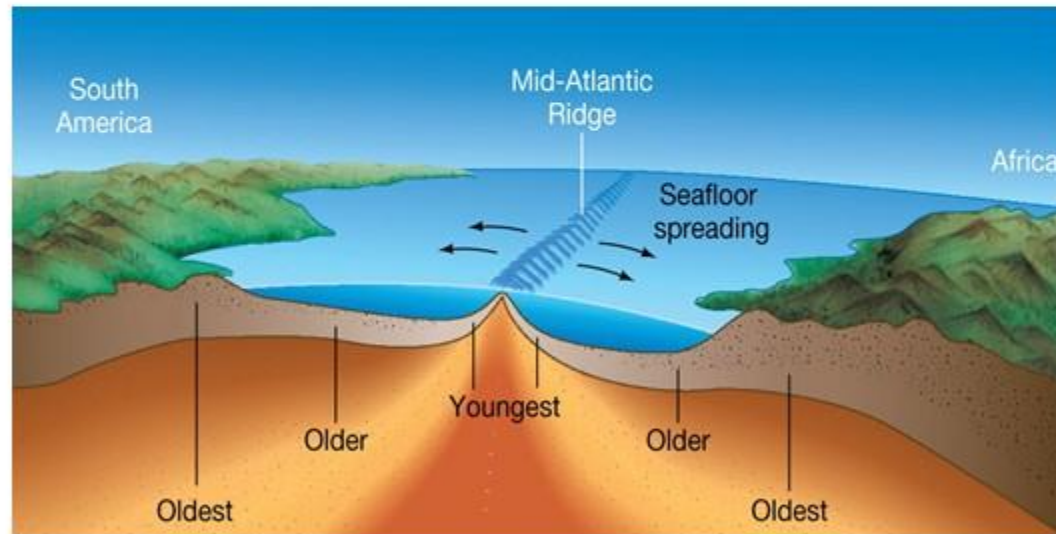
MINOR RELIEF FEATURES



Apart from the above mentioned major relief features of the ocean floor, some minor but significant features predominate in different parts of the oceans.

1

MID-OCEANIC RIDGES



1

A mid-oceanic ridge is composed of two chains of mountains separated by a large depression.

2

The mountain ranges can have peaks as high as 2,500 m and some even reach above the ocean's surface.

3

Iceland, part of the mid-Atlantic Ridge, is an example.



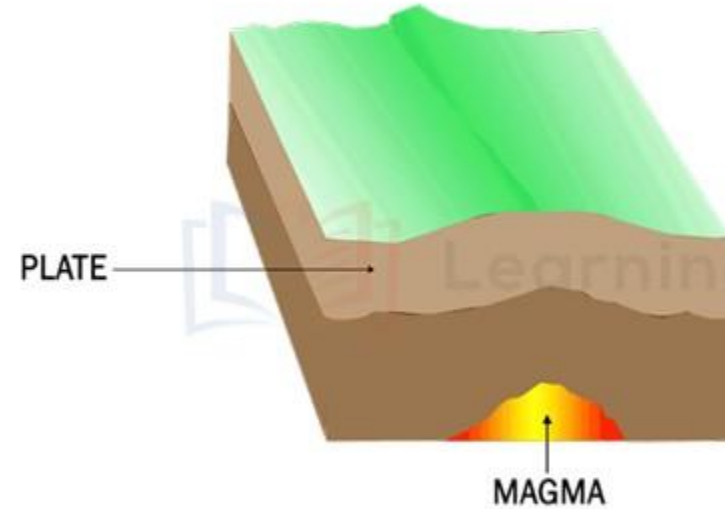
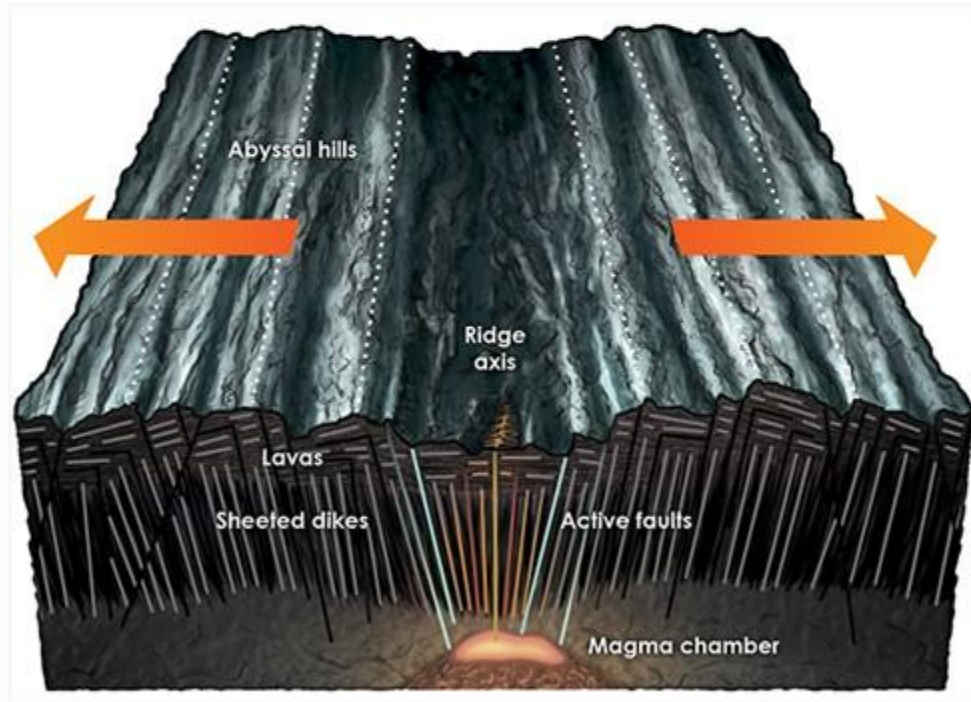
OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES

1

MID-OCEANIC RIDGES





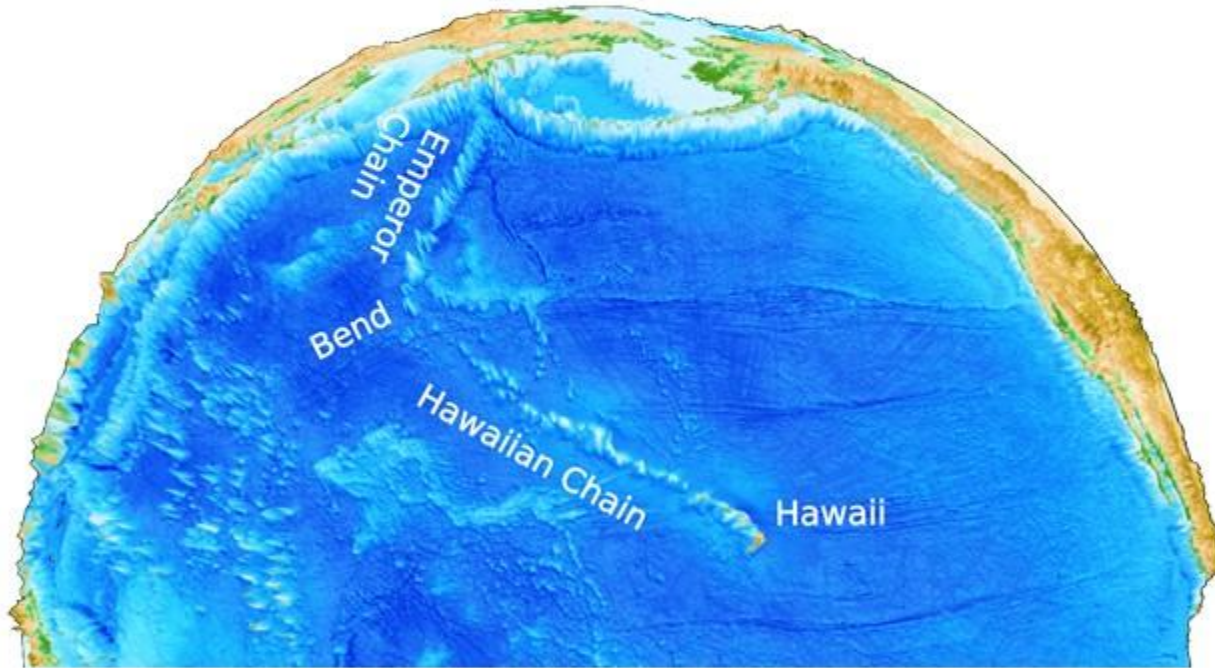
OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES

2

SEAMOUNT



HAWAIIAN-EMPEROR SEAMOUNT CHAIN

- 1 It is a mountain with pointed summits, rising from the seafloor that does not reach the surface of the ocean.
- 2 Seamounts are **volcanic** in origin. These can be 3,000-4,500 m tall.
- 3 The **Emperor seamount**, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example.



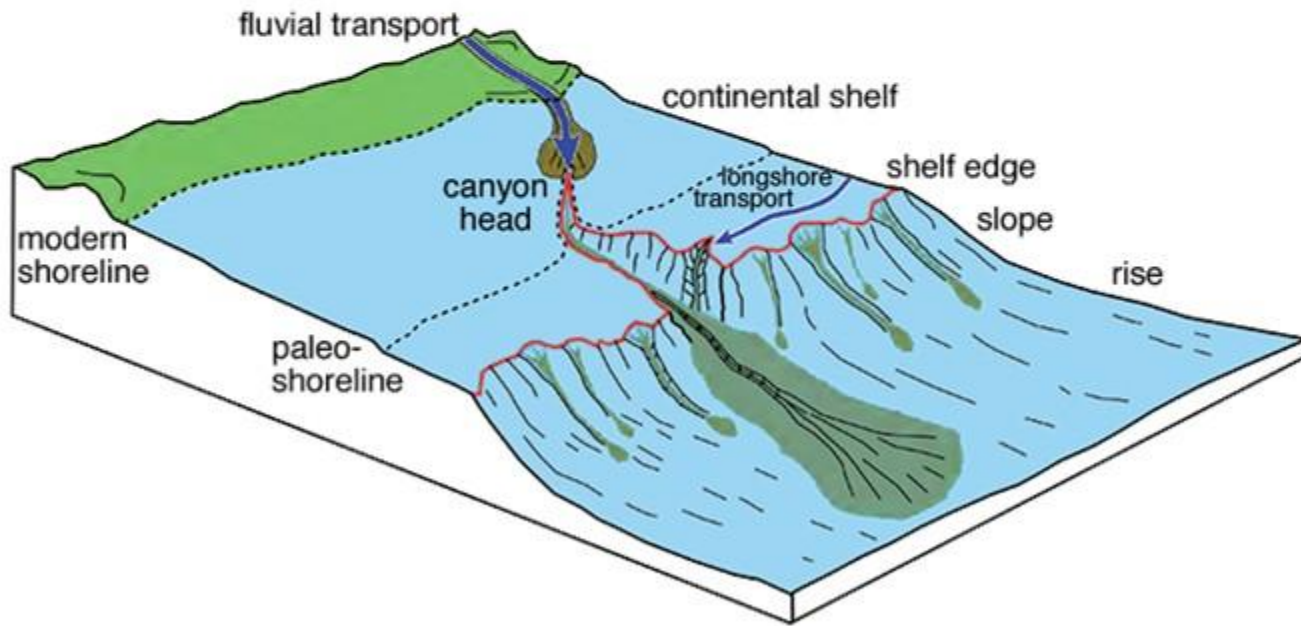
OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES

3

SUBMARINE CANYONS



- 1 These are **deep valleys**, some comparable to the Grand Canyon of the Colorado river.
- 2 They are sometimes found cutting across the continental shelves and slopes, often extending from the mouths of large rivers.
- 3 The **Hudson Canyon** is the best known canyon in the world.



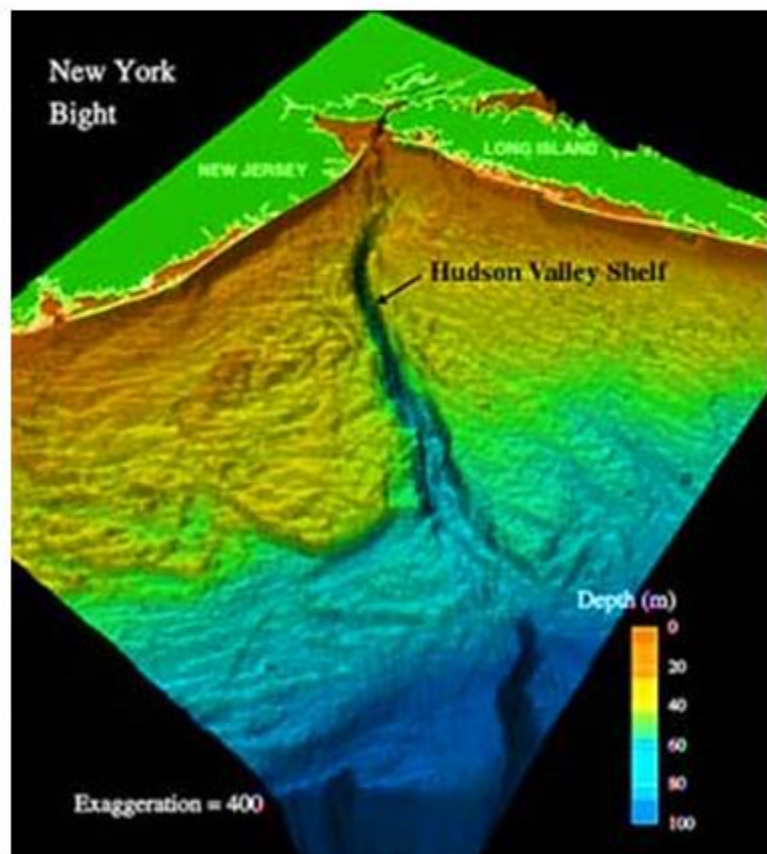
OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES

3

SUBMARINE CANYONS



HUDSON CANYON

- *The Hudson Canyon is a submarine canyon that begins from the shallow outlet of the estuary at the mouth of the Hudson River.*
- *It extends out over 640 km seaward across the continental shelf finally connecting to the deep ocean basin at a depth of 3 to 4 km below sea level.*
- *Its walls rise three-quarters of a mile from the canyon floor making it comparable to the Grand Canyon.*



OCEANOGRAPHY

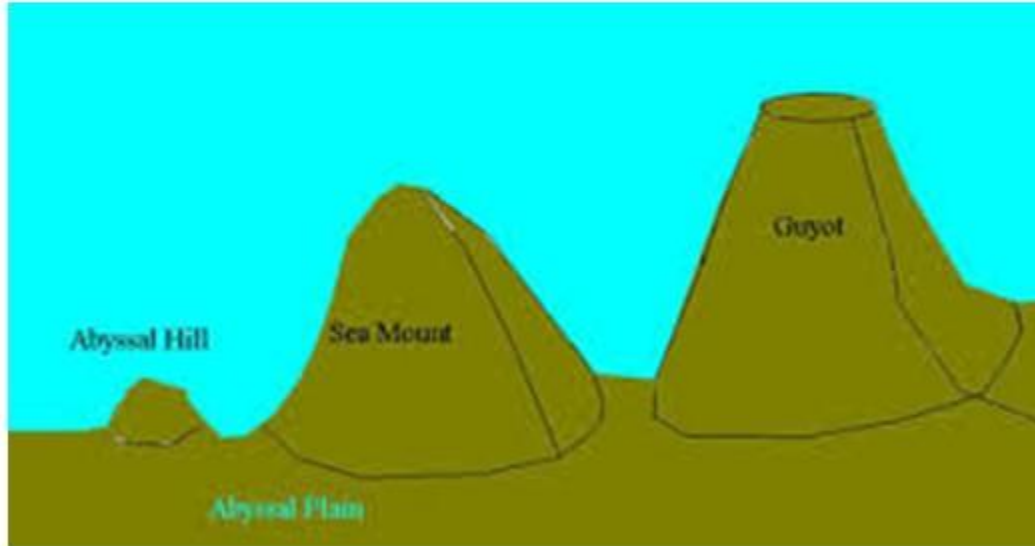
DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES



4

GUYOTS



- 1 It is a flat topped seamount.
- 2 They show evidences of gradual subsidence through stages to become flat topped submerged mountains.
- 3 It is estimated that more than 10,000 seamounts and guyots exist in the Pacific Ocean alone.



OCEANOGRAPHY

DIVISIONS OF THE OCEAN FLOORS

MINOR RELIEF FEATURES

5

ATOLL



- 1 This is a ring of narrow growing corals generally formed around an island or in elliptical form on a submarine platform.
- 2 It may be a part of the sea (lagoon), or sometimes form enclosing a body of fresh or brackish water.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9